Projective Transformations

Prof. Ziyun (Claude) Wang
Department of Electrical and Computer Engineering
Spring 2026

Announcement

- HW 1 out this week, due in two weeks
- Two parts, all due on GradeScope
- All office hours this week will be in person at normal time

Recap: Full Camera Matrix

General intrinsics (principal point, pixel scaling, optional skew)

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

- f_x, f_y: focal length measured in *pixels* (includes pixel size / aspect ratio)
- (c_x, c_y): **principal point** (projection of optical axis) in pixel coordinates
- s: skew (often ≈ 0)

Recap: projecting world points

Now assume the point is given in a **world frame** as $\boldsymbol{X}_w = (X_w, Y_w, Z_w)^\top$.

Extrinsics (camera pose)

The camera-frame coordinates are

$$\boldsymbol{X}_c = R_w^c \, \boldsymbol{X}_w + \boldsymbol{t}_w^c, \qquad R_w^c \in SO(3), \; \boldsymbol{t}_w^c \in \mathbb{R}^3.$$

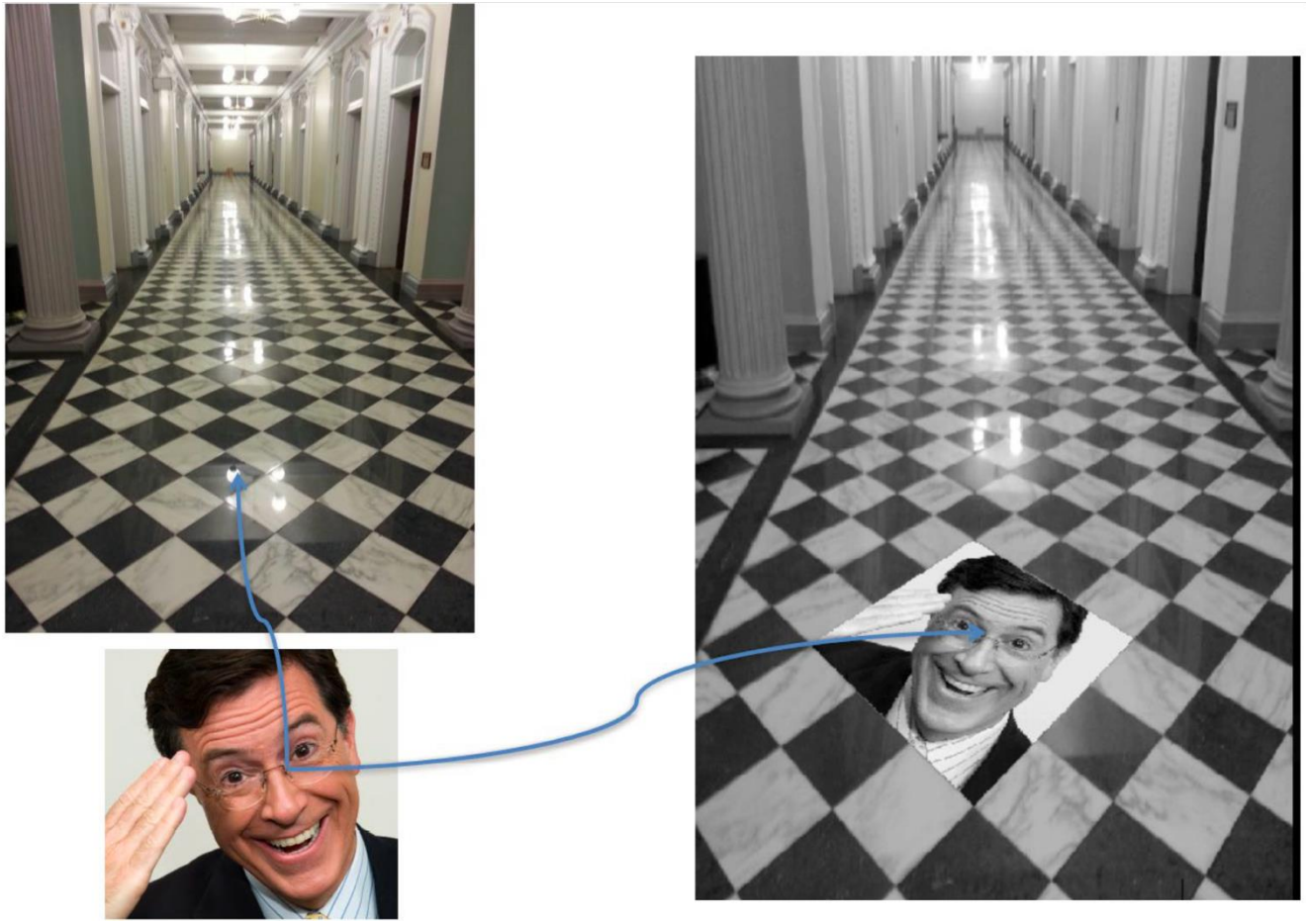
In homogeneous form:

$$\begin{bmatrix} \boldsymbol{X}_c \\ 1 \end{bmatrix} = \begin{bmatrix} R_w^c & \boldsymbol{t}_w^c \\ \boldsymbol{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{X}_w \\ 1 \end{bmatrix}.$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \boxed{\underbrace{\begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix}}_{[R\,|\,\boldsymbol{t}]}} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}.$$

3 x 4 Projection Matrix

How do we do this mapping?



Homography / Collineation / Projective Transformation

Projective Transformation / Homography

A **projective transformation** is any invertible matrix transformation

$$\mathbb{P}^2 \to \mathbb{P}^2.$$

- A projective transformation $\mathbf{H}$ maps a point $\boldsymbol{p}$ to $\boldsymbol{p}' \sim \mathbf{H}\boldsymbol{p}$.
- Invertibility means $\det(\mathbf{H}) \neq 0$ and that there exists $\lambda \neq 0$ such that

$$\lambda \boldsymbol{p}' = \mathbf{H}\boldsymbol{p}.$$

- We will write either $\boldsymbol{p}' \sim \mathbf{H}\boldsymbol{p}$ or $\lambda \boldsymbol{p}' = \mathbf{H}\boldsymbol{p}$.

What does this mean geometrically?

Example: homography

Consider the (affine) homography

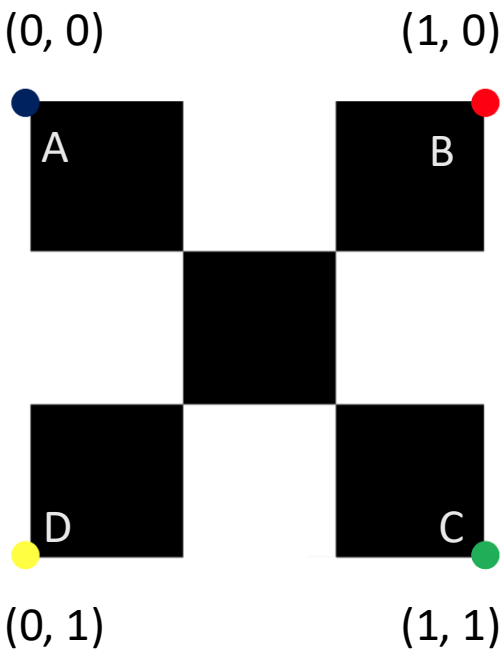
$$\mathbf{H} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

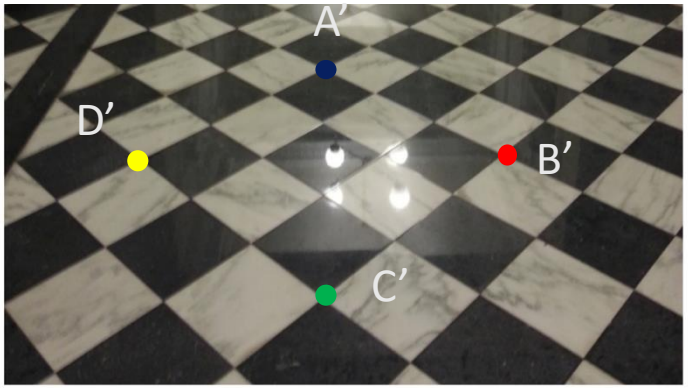
Take a point $\boldsymbol{p} = (1, 1)$, i.e.

$$\boldsymbol{p}_h = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

$$\boldsymbol{p}'_h \sim \mathbf{H}\boldsymbol{p}_h = \begin{bmatrix} 2 \cdot 1 + 0 \cdot 1 + 1 \\ 0 \cdot 1 + 1 \cdot 1 + 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} \Rightarrow \boldsymbol{p}' = (3, 3).$$

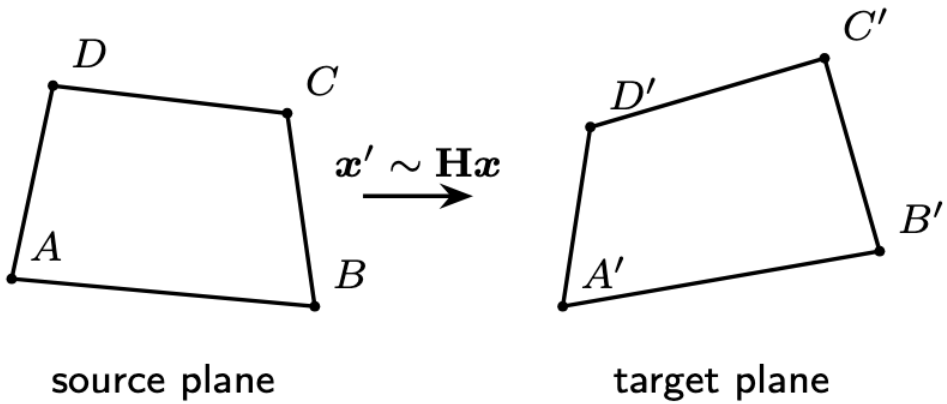
Example: square to tile





What a homography can do

- warp shapes while preserving straightness.



What is preserved?

Preserved by any collineation:

- straight lines (collinearity)
- incidence (point-on-line)
- cross-ratio of four collinear points

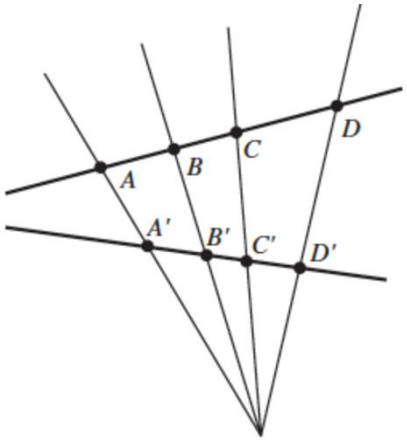
Not preserved in general:

- Euclidean lengths and angles
- parallelism
- ratios of distances (unless additional structure exists)

Cross Ratio

For four collinear points A, B, C, D with affine coordinates a, b, c, d:

$$(A, B; C, D) = \frac{(c-a)(d-b)}{(c-b)(d-a)}$$



Affine coordinate means each point corresponds to a scalar on a line

Degrees of freedom of a homography

- What is Degrees of Freedom (DOF) of an object?
  - Intuition: total number of independent need to specify a unique object
  - Need to remove redundancy

Example: a 2D line

A line is represented by $\boldsymbol{\ell} = (a, b, c)^\top \neq \mathbf{0}$ with equation $\boldsymbol{\ell}^\top \boldsymbol{x} = 0$. But $\boldsymbol{\ell}$ and $\lambda\boldsymbol{\ell}$ define the same line, so

$$\text{DoF(line in } \mathbb{P}^2) = 3 - 1 = 2.$$

Minus 1 “irrelevant overall scale”

Degrees of freedom of a homography

General rule

If an object is parameterized by a nonzero vector in $\mathbb{R}^n$ and $\boldsymbol{v} \sim \lambda \boldsymbol{v}$ for any $\lambda \neq 0$, then the degrees of freedom are

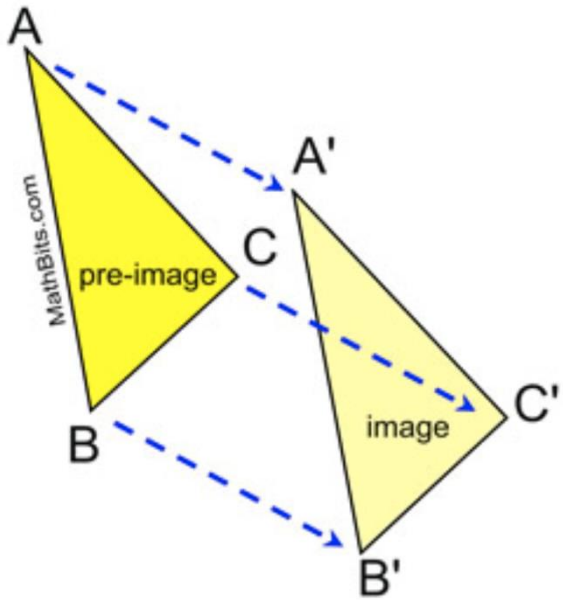
$$\text{DoF} = n - 1.$$

DOF of transforms

Translation (2 DoF)

$$\bar{\boldsymbol{x}}' = \underbrace{\begin{bmatrix} \mathbf{I} & \boldsymbol{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}}_{T_{\text{trans}}} \bar{\boldsymbol{x}}.$$

Preserves: distances and angles (just shifts the origin).



DOF of Transforms

Rotation, 3 DOF

$$\boldsymbol{x} \mapsto \boldsymbol{x}' = R\boldsymbol{x}, \qquad R \in \mathbb{R}^{3\times 3}.$$

$$\bar{\boldsymbol{x}}' = \underbrace{\begin{bmatrix} R & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}}_{T_{\mathsf{rot}}} \bar{\boldsymbol{x}}, \qquad R \in \mathbb{R}^{3\times 3}.$$

DOF of Transforms: rigid

A 2D rigid motion (rotation + translation) can be written in homogeneous coordinates as

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R(\theta) & t \\ 0^\top & 1 \end{bmatrix}}_{H_E} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \qquad R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \; t \in^2 .$$

Degrees of freedom

θ (1) + translation t (2) $\Rightarrow$ **3 DOF**.

What it preserves

Lengths, angles, and areas. (Also preserves orientation.)

DOF of Transforms: similarity

A similarity adds a global scale $s > 0$:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} sR(\theta) & t \\ 0^\top & 1 \end{bmatrix}}_{H_S} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \qquad s > 0.$$

Degrees of freedom

Scale s (1) + rotation θ (1) + translation t (2) $\Rightarrow$ **4 DOF**.

What it preserves

Angles and shapes up to a global scale. (Also preserves parallel lines.)

DOF of Transforms: affine

An affine transform is the most general linear map plus translation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} A & t \\ 0^\top & 1 \end{bmatrix}}_{H_A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \qquad A \in^{2\times 2},\ \det(A) \neq 0.$$

Degrees of freedom

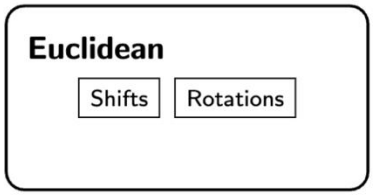
A has 4 parameters, t has 2 $\Rightarrow$ **6 DOF**.

What it preserves

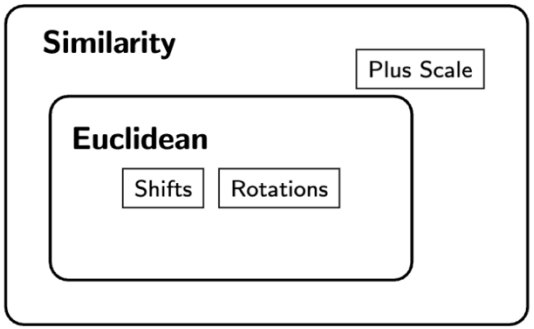
Parallelism and ratios of lengths along any fixed direction (e.g., along a line).

Hierarchy of transforms

- **Euclidean (Rigid)**
  Preserves: Lengths, Areas, Angles.

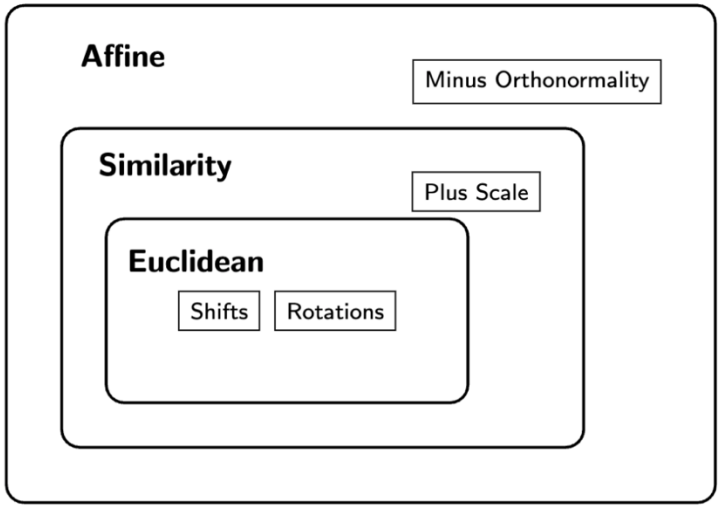


Hierarchy of transforms



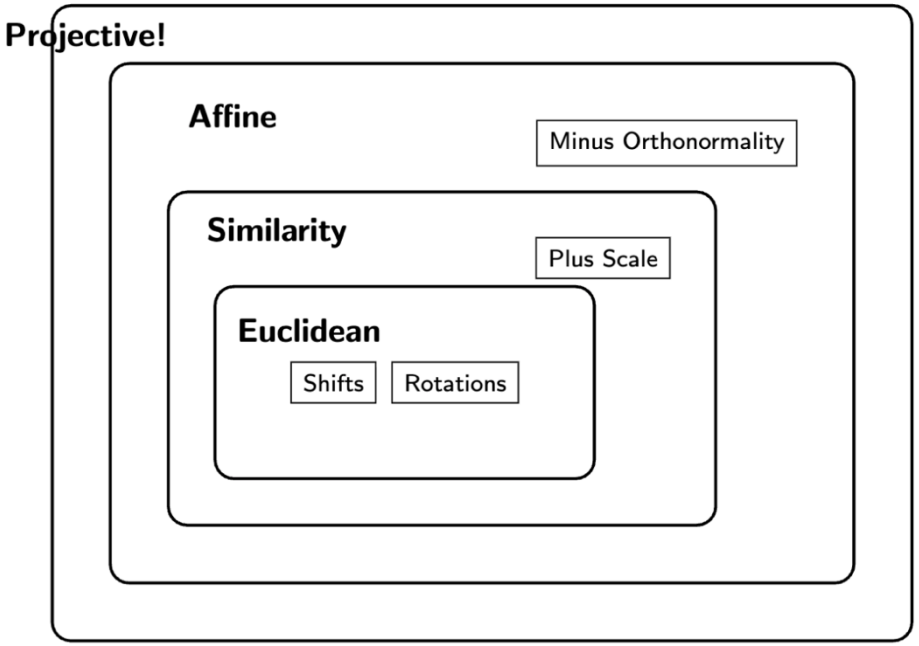
- **Euclidean (Rigid)**
  Preserves: Lengths, Areas, Angles.
- **Similarity**
  Preserves: Angles, Parallelism.

Hierarchy of transforms



- **Euclidean (Rigid)**
  Preserves: Lengths, Areas, Angles.
- **Similarity**
  Preserves: Angles, Parallelism.
- **Affine**
  Preserves: Parallelism, Ratio of Areas.

Hierarchy of transforms



- **Euclidean (Rigid)**
  Preserves: Lengths, Areas, Angles.
- **Similarity**
  Preserves: Angles, Parallelism.
- **Affine**
  Preserves: Parallelism, Ratio of Areas.
- **Projective (Homography)**
  Preserves: Collinearity, Cross-ratio.

Why does DOF matter?

DOF of a transform decides the minimum number of independent equations that we need to solve for a transform

Line has 2 DOF -> two points minimal to solved for a line

Constraints from correspondence

Write a point in homogeneous coordinates as

$$\boldsymbol{x} = (x, y, 1)^\top, \qquad \boldsymbol{x}' \sim \mathbf{H}\boldsymbol{x}.$$

Expand $\mathbf{H}\boldsymbol{x}$ entry-by-entry

Let

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}.$$

Then

$$\tilde{\boldsymbol{x}}' = \mathbf{H}\boldsymbol{x} = \begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = \begin{bmatrix} h_{11}x + h_{12}y + h_{13} \\ h_{21}x + h_{22}y + h_{23} \\ h_{31}x + h_{32}y + h_{33} \end{bmatrix}.$$

If $w' \neq 0$, dehomogenize to get (x', y'):

$$x' = \frac{u'}{w'}, \qquad y' = \frac{v'}{w'}, \qquad \boldsymbol{x}' = (x', y', 1)^\top.$$

Constraints from correspondence

Given N correspondences $(\boldsymbol{x}_i \leftrightarrow \boldsymbol{x}'_i)$ with $\boldsymbol{x}'_i \sim \mathbf{H}\boldsymbol{x}_i$.
Write $\boldsymbol{x} = (x, y, 1)^\top$ and $\boldsymbol{x}' = (u, v, 1)^\top$. In inhomogeneous form:

$$u = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}},$$
$$v = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}.$$

Each correspondence gives us two equations

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -ux & -uy & -u \\ 0 & 0 & 0 & x & y & 1 & -vx & -vy & -v \end{bmatrix} \boldsymbol{h} = \mathbf{0},$$

$$\boldsymbol{h} = \begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{21} & h_{22} & h_{23} & h_{31} & h_{32} & h_{33} \end{bmatrix}^\top.$$

Direct Linear Transform

Per-point block

For correspondence i with $(x_i, y_i) \leftrightarrow (u_i, v_i)$, define

$$\mathbf{A}_i = \begin{bmatrix} x_i & y_i & 1 & 0 & 0 & 0 & -u_i x_i & -u_i y_i & -u_i \\ 0 & 0 & 0 & x_i & y_i & 1 & -v_i x_i & -v_i y_i & -v_i \end{bmatrix} \in \mathbb{R}^{2\times 9}.$$

Stack all points

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_N \end{bmatrix} \in \mathbb{R}^{2N\times 9}, \qquad \mathbf{A}\boldsymbol{h} = \mathbf{0}.$$

Since $\boldsymbol{h}$ is defined up to scale, solve

$$\min_{\|\boldsymbol{h}\|=1} \|\mathbf{A}\boldsymbol{h}\|^2 \quad \Rightarrow \quad \boldsymbol{h} = \text{right singular vector of } \mathbf{A} \text{ with smallest singular value.}$$

Solving homogenous equations with SVD

For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the singular value decomposition (SVD) is

$$\mathbf{A} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^\top,$$

where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ are orthonormal ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}$, $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$)
- $\mathbf{\Sigma}$ is diagonal with nonnegative singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$

Why SVD solves Ah = 0?

In DLT we want a nonzero solution $\boldsymbol{h}$ to the homogeneous system $\mathbf{A}\boldsymbol{h} = \mathbf{0}$. With noise, we solve the constrained least-squares problem

$$\min_{\|\boldsymbol{h}\|=1} \|\mathbf{A}\boldsymbol{h}\|^2.$$

Using $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ and letting $\boldsymbol{h} = \mathbf{V}\boldsymbol{z}$,

$$\|\mathbf{A}\boldsymbol{h}\|^2 = \|\boldsymbol{\Sigma}\boldsymbol{z}\|^2.$$

So the minimizer is $\boldsymbol{z} = \boldsymbol{e}_n$ (the last basis vector), i.e.

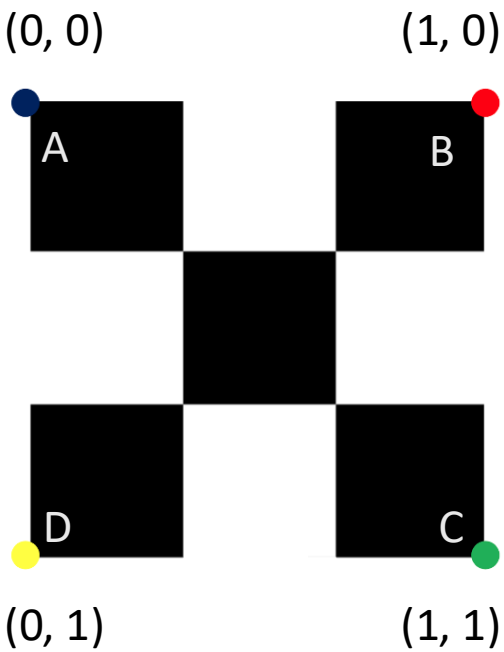
$$\boldsymbol{h} = \text{last column of } \mathbf{V} \text{ (smallest singular value).}$$

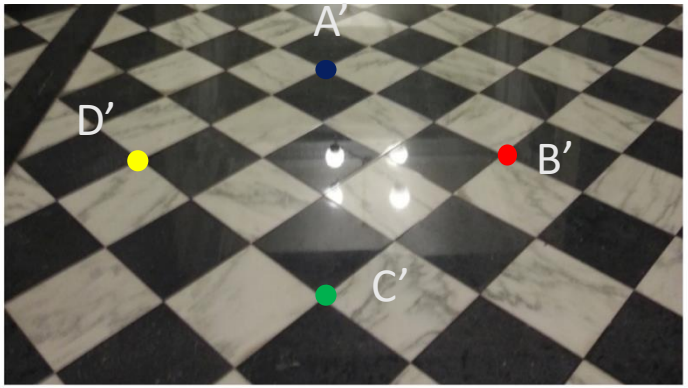
1. Each column of V is always a unit vector, and H is up to a scale
2. If fewer than 4 points, multiple solutions

Compute homography from 4 points

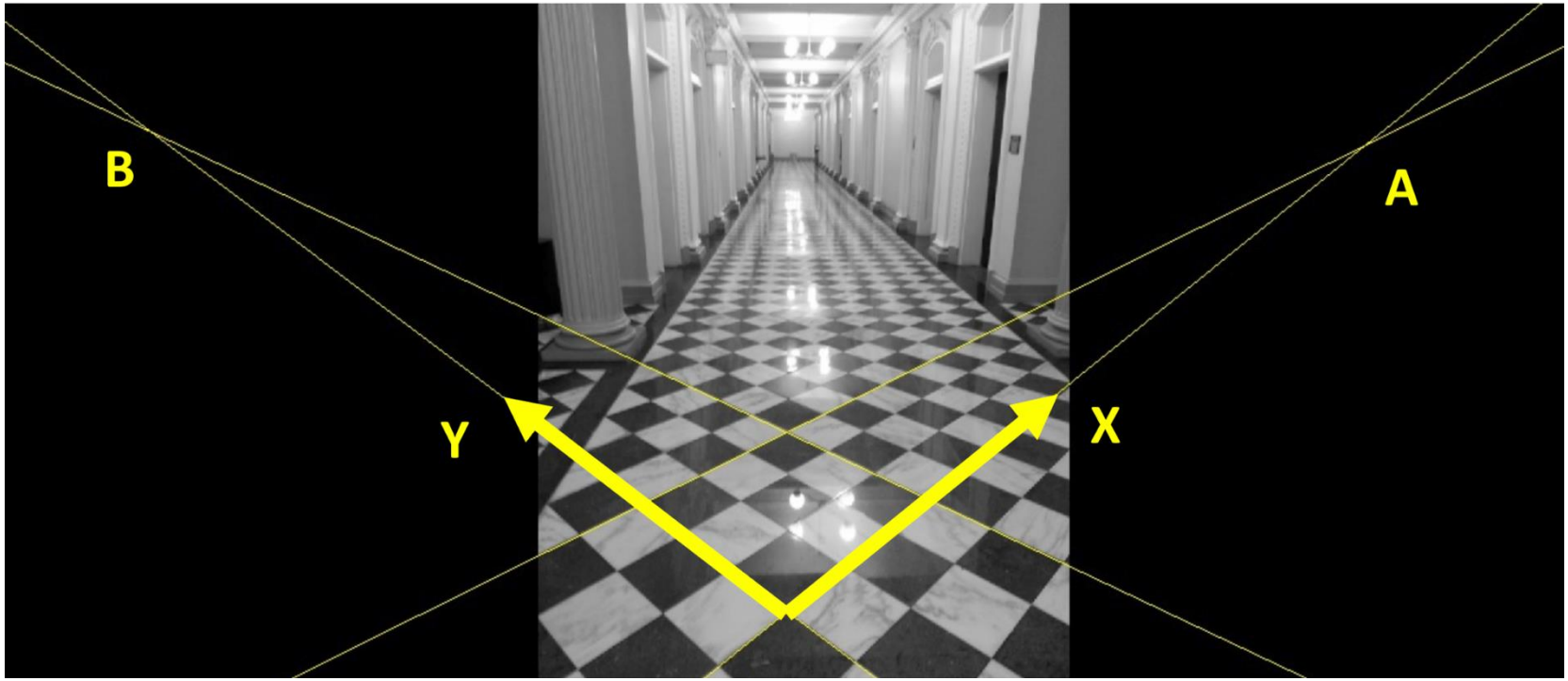
- We need 4 independent points to solve
- Solve homogeneous equations with **SVD**
- Need **RANSAC** to improve robustness
  - Will come back to RANSAC

Simpler way to solve for H





Homography and Vanishing Points



Homography and Vanishing Points

Let a plane-to-image homography be written in columns:

$$\mathbf{H}_{W\to I} = \begin{bmatrix} | & | & | \\ \boldsymbol{h}_1 & \boldsymbol{h}_2 & \boldsymbol{h}_3 \\ | & | & | \end{bmatrix}.$$

Points at infinity on the plane

In plane coordinates $\mathbf{X}_\pi = (X, Y, 1)^\top$, the directions of the plane axes are the points at infinity

$$\mathbf{v}_x = (1, 0, 0)^\top, \qquad \mathbf{v}_y = (0, 1, 0)^\top.$$

Their images are

$$\boldsymbol{x}_x \sim \mathbf{H}\mathbf{v}_x = \boldsymbol{h}_1, \qquad \boldsymbol{x}_y \sim \mathbf{H}\mathbf{v}_y = \boldsymbol{h}_2,$$

First two columns are just scaled vanishing points, and last column is just the origin

Homography and Vanishing Points

In plane coordinates, the origin is

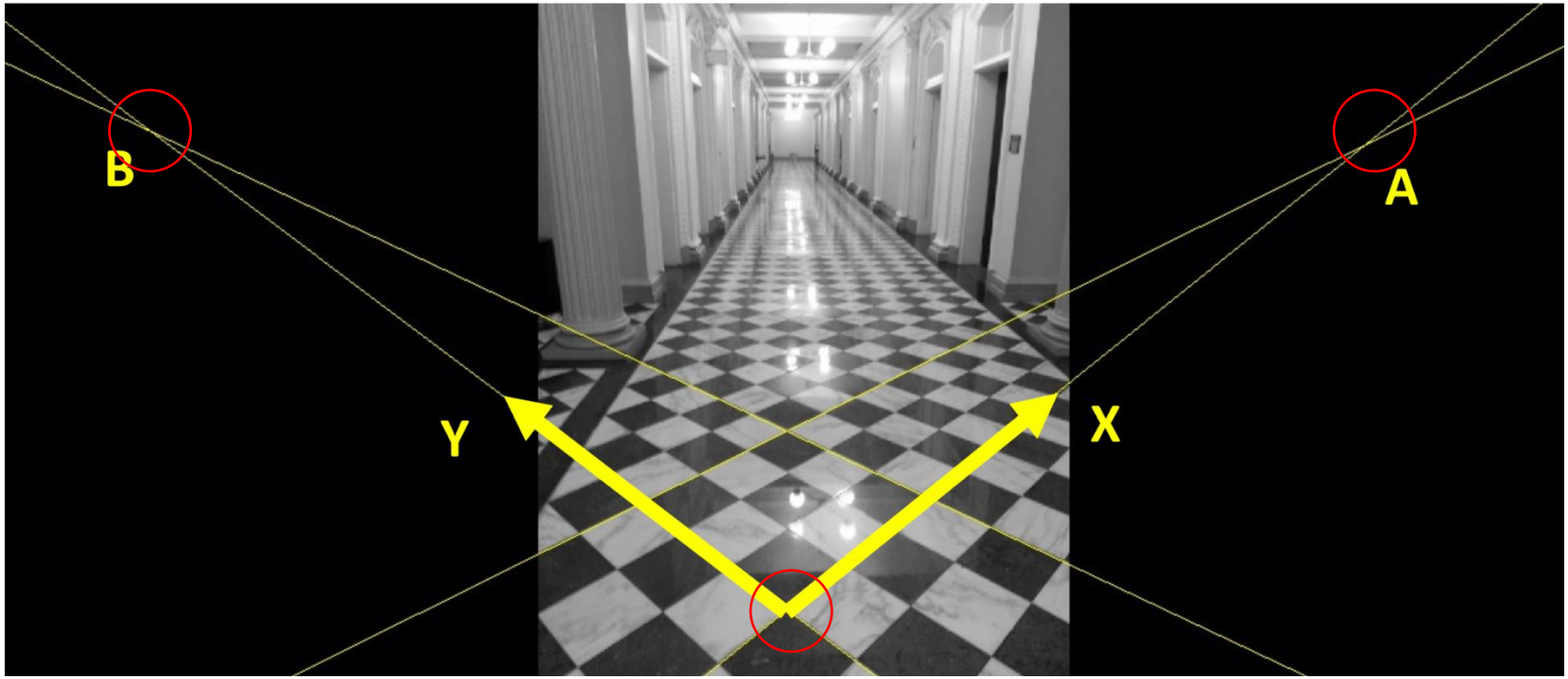
$$\mathbf{X}_{\pi,0} = (0, 0, 1)^\top.$$

Under the plane-to-image homography,

$$\boldsymbol{x}_0 \sim \mathbf{H}_{W\to I}\,\mathbf{X}_{\pi,0} = \mathbf{H}_{W\to I}\begin{bmatrix}0\\0\\1\end{bmatrix} = \boldsymbol{h}_3.$$

The first two columns are just scaled vanishing points, and the last column is just the origin
If we want the full homography, need 4th point

Homography and Vanishing Points



Fix scale with a fourth point

Assume the same mapping also sends

$$\boldsymbol{x}_4 = (1,1,1)^\top \;\mapsto\; \boldsymbol{d} \in \mathbb{P}^2.$$

Then the projective relation implies there exists $\lambda \neq 0$ such that

$$\lambda \boldsymbol{d} = \mathbf{H}\boldsymbol{x}_4 = \begin{pmatrix} \alpha\boldsymbol{a} & \beta\boldsymbol{b} & \gamma\boldsymbol{c} \end{pmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha\boldsymbol{a} + \beta\boldsymbol{b} + \gamma\boldsymbol{c}.$$

Fix the scale and solve a 3×3 system

Because points in $\mathbb{P}^2$ are defined up to scale, we can choose a representative of $\boldsymbol{d}$ so that $\lambda = 1$. Then

$$\underbrace{\begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} \end{bmatrix}}_{\in \mathbb{R}^{3\times 3}} \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \boldsymbol{d}.$$

Homography in single view projection

A planar projection is a homography

Why is planar projection homographic?

Start from the pinhole projection of a 3D world point $\mathbf{X} = (X, Y, Z, 1)^\top$:

$$\boldsymbol{x} \sim \mathbf{P}\,\mathbf{X}, \qquad \mathbf{P} = \boxed{\mathbf{K}\,[\mathbf{R} \mid \mathbf{t}] \in \mathbb{R}^{3\times 4}.}$$

Pose from camera to a canonical frame on the plane

Write the rotation matrix in columns:

$$\mathbf{R} = \begin{bmatrix} | & | & | \\ \boldsymbol{r}_1 & \boldsymbol{r}_2 & \boldsymbol{r}_3 \\ | & | & | \end{bmatrix}.$$

Then

$$\mathbf{P} = \mathbf{K}[\mathbf{R} \mid \mathbf{t}] = \begin{bmatrix} | & | & | & | \\ \mathbf{K}\boldsymbol{r}_1 & \mathbf{K}\boldsymbol{r}_2 & \mathbf{K}\boldsymbol{r}_3 & \mathbf{K}\mathbf{t} \\ | & | & | & | \end{bmatrix}.$$

Why is planar projection homographic?

If the world is defined so the plane is $Z = 0$, then points on the plane have

$$\mathbf{X}_\pi = (X, Y, 0, 1)^\top.$$

Writing $\mathbf{P}$ in columns:

$$\mathbf{P} = \begin{bmatrix} | & | & | & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_3 & \mathbf{p}_4 \\ | & | & | & | \end{bmatrix}, \qquad \boldsymbol{x} \sim \mathbf{P}\,\mathbf{X}_\pi = X\,\mathbf{p}_1 + Y\,\mathbf{p}_2 + \mathbf{p}_4.$$

Therefore

$$\boldsymbol{x} \sim \underbrace{\begin{bmatrix} | & | & | \\ \mathbf{p}_1 & \mathbf{p}_2 & \mathbf{p}_4 \\ | & | & | \end{bmatrix}}_{\mathbf{H} \in \mathbb{R}^{3\times 3}} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix},$$

so the full 3×4 projection reduces to a 3×3 mapping in homogeneous coordinates: a **homography**.

Summary: planar projection

- Any point on a plane will share the same 3 by 3 projection
- A general projection degenerates to homography for planar scenes

Homography between two views

- When the two views are looking at a planar scene
- When the two views are related by a pure rotation

Case 1: planar scenes

Pick coordinates on the plane (a 2D coordinate system), represented in homogeneous form as

$$\mathbf{X}_\pi = (X, Y, 1)^\top \in \mathbb{P}^2.$$

Each view is a plane-to-image homography

For a planar surface, each camera maps plane coordinates to the image by a 3×3 homography:

$$\boldsymbol{x}_1 \sim \mathbf{H}_1 \, \mathbf{X}_\pi, \qquad \boldsymbol{x}_2 \sim \mathbf{H}_2 \, \mathbf{X}_\pi.$$

Eliminate $\mathbf{X}_\pi$:

$$\mathbf{X}_\pi \sim \mathbf{H}_1^{-1} \, \boldsymbol{x}_1 \quad \Rightarrow \quad \boldsymbol{x}_2 \sim \mathbf{H}_2 \, \mathbf{H}_1^{-1} \, \boldsymbol{x}_1.$$

Equivalently, mapping the other direction gives $\boldsymbol{x}_1 \sim \mathbf{H}_1 \, \mathbf{H}_2^{-1} \, \boldsymbol{x}_2$.

Case 2: pure rotations

If the camera only rotates (no translation), then

$$\mathbf{P} = \mathbf{K}[\mathbf{R} \mid \mathbf{0}].$$

Consider two views of a static scene with the same camera center:

$$\boldsymbol{x}_1 \sim \mathbf{K}_1 \, [\mathbf{I} \mid \mathbf{0}] \, \mathbf{X}, \qquad \boldsymbol{x}_2 \sim \mathbf{K}_2 \, [\mathbf{R} \mid \mathbf{0}] \, \mathbf{X}.$$

From the first equation, $\mathbf{X}$ is proportional to $[\mathbf{K}_1^{-1}\boldsymbol{x}_1;\ 1]$ (up to depth), so eliminating $\mathbf{X}$ gives

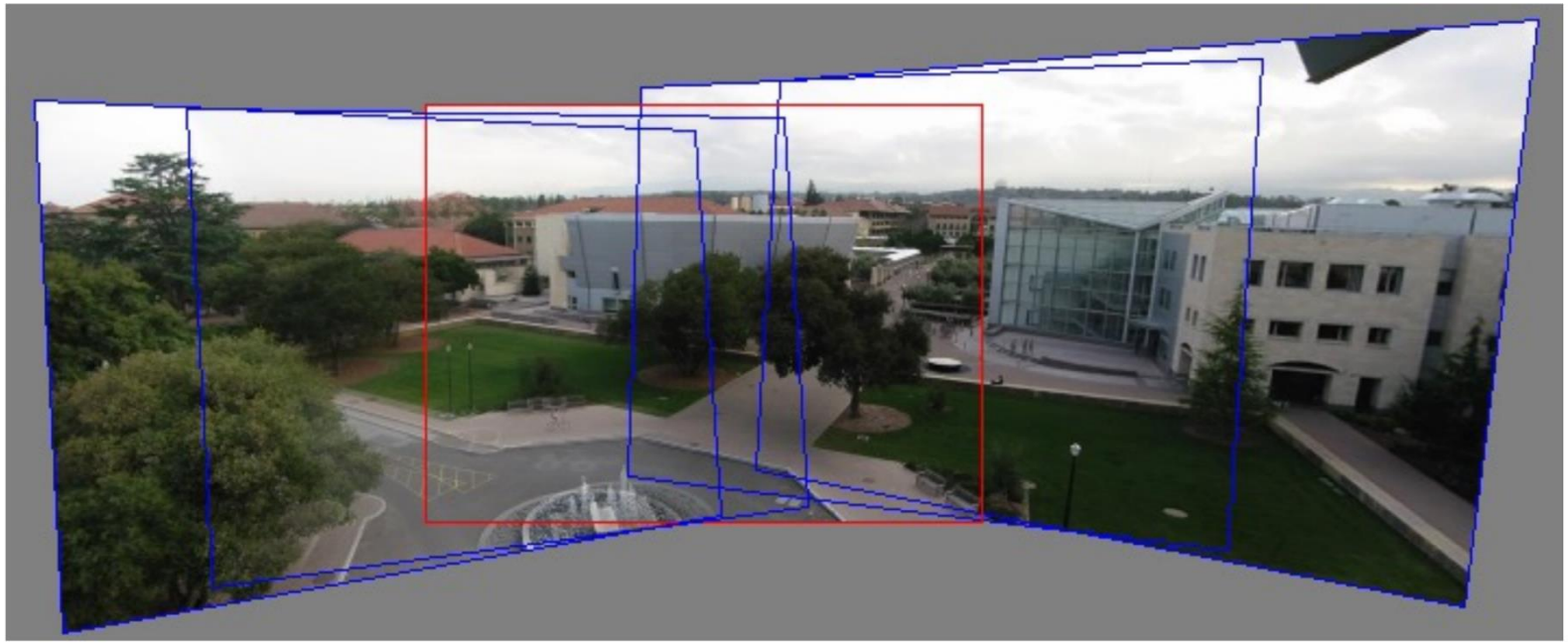
$$\boldsymbol{x}_2 \sim \mathbf{K}_2 \, \mathbf{R} \, \mathbf{K}_1^{-1} \, \boldsymbol{x}_1.$$

Result

The mapping between the two images is a homography:

$$\boldsymbol{x}_2 \sim \mathbf{H} \, \boldsymbol{x}_1, \qquad \mathbf{H} = \mathbf{K}_2 \, \mathbf{R} \, \mathbf{K}_1^{-1}.$$

Panoramic photos



Application: Metric Measurement in World